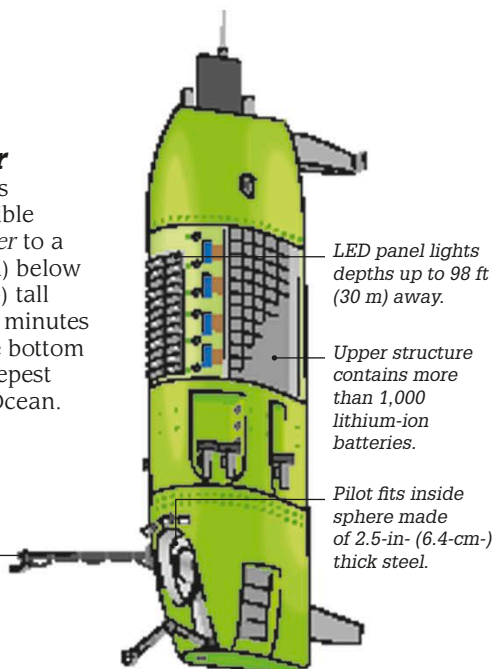


2012

Deepsea Challenger

Canadian film director James Cameron piloted a submersible called the *Deepsea Challenger* to a depth of 35,787 ft (10,908 m) below sea level. The 24-ft- (7.3-m-) tall craft took two hours and 37 minutes to dive from sea level to the bottom of Challenger Deep—the deepest known point in the Pacific Ocean.

Mobile boom carries a powerful spotlight and 3-D cameras.



2012

Higgs boson

In July, scientists working at the Large Hadron Collider discovered the Higgs boson. This elusive particle proved the existence of the Higgs field, which had been predicted but never proven in the past. The existence of the Higgs field solves the mystery of how individual subatomic particles have mass.

A
smashing
time
See pages
240–241

2014

Rosetta reaches comet

After a ten-and-a-half year journey from Earth, the European Space Agency's Rosetta space probe went into orbit around Comet 67P/Churyumov-Gerasimenko. The probe launched a lander named Philae, which made the first ever soft landing on a comet and sent back measurements from its surface.



Artist's impression of the Philae lander on the surface of Comet 67P

2015



2012

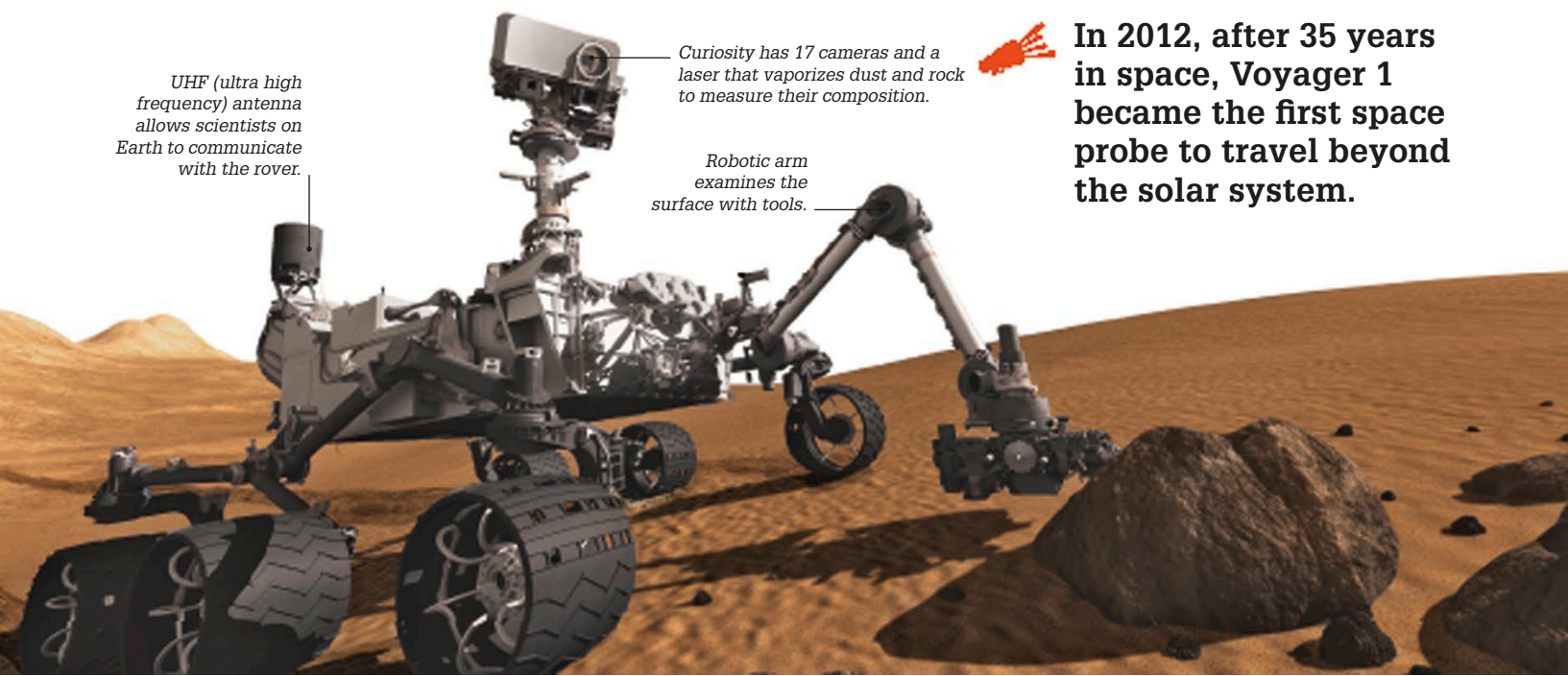
Curiosity on Mars

NASA's Curiosity rover arrived on Mars after a 350-million-mile (563-million-km) journey. The 1,982-lb (899-kg) rover was 78 times the weight of the Sojourner rover from 1997. Curiosity also included 176 lb (80 kg) of scientific instruments for studying Martian weather, soil, rocks, and radiation, and looking for potential water sources as well as signs of life in the past.

UHF (ultra high frequency) antenna allows scientists on Earth to communicate with the rover.

Curiosity has 17 cameras and a laser that vaporizes dust and rock to measure their composition.

Robotic arm examines the surface with tools.



2014

Nanomotor

A team led by American inventor Dr. Donglei Fan at the University of Texas, produced the world's smallest motor (see p.245). This nanomotor is about 500 times smaller than a grain of salt. Future models could deliver drugs directly into individual body cells.



In 2012, after 35 years in space, Voyager 1 became the first space probe to travel beyond the solar system.